

JAVA LAB RECORD

Experiment No.	06
Date	23/09/26
Title	Implementation of LinkedList Methods in Java

Objective / Aim

To study and implement commonly used methods of the Java LinkedList class, including adding, accessing, removing, queue operations, and retrieving the first and last elements.

System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Keyboard and mouse
- Monitor/display

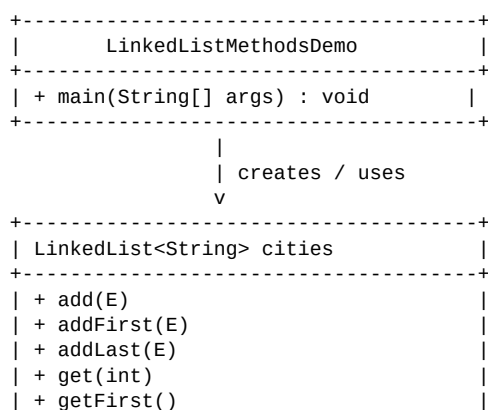
Software Requirements

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK): 17 or later
- IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program.
2. Import the LinkedList class from java.util.
3. Create a LinkedList named cities to store String values.
4. Add city names using add(E e).
5. Insert a city at the beginning using addFirst(E e).
6. Insert a city at the end using addLast(E e).
7. Access an element at a particular position using get(index).
8. Retrieve the first and last elements using getFirst() and getLast().
9. Remove an element using its index with remove(index).
10. Remove a specific city using remove(Object).
11. Remove and return the first element using removeFirst().
12. Remove and return the last element using removeLast().
13. Insert an element using offer(E e), which treats the LinkedList as a queue.
14. View the head element using peek() without removing it.
15. Remove and return the head element using poll().
16. Stop the program.

UML Class Diagram



```
| + getLast() |
| + remove(int) |
| + remove(Object) |
| + removeFirst() |
| + removeLast() |
| + offer(E) |
| + peek() |
| + poll() |
+-----+
```

Source Code

```
import java.util.LinkedList;

class LinkedListMethodsDemo {

    public static void main(String[] args) {

        // Creating a LinkedList
        LinkedList<String> cities = new LinkedList<>();

        // add(E e)
        cities.add("Hyderabad");
        cities.add("Chennai");
        cities.add("Mumbai");

        System.out.println("After add(): " + cities);

        // addFirst(E e)
        cities.addFirst("Delhi");
        System.out.println("After addFirst(): " + cities);

        // addLast(E e)
        cities.addLast("Kolkata");
        System.out.println("After addLast(): " + cities);

        // get(int index)
        System.out.println("Element at index 2: " + cities.get(2));

        // getFirst()
        System.out.println("First element: " + cities.getFirst());

        // getLast()
        System.out.println("Last element: " + cities.getLast());

        // remove(int index)
        cities.remove(2);
        System.out.println("After remove(index): " + cities);

        // remove(Object o)
        cities.remove("Chennai");
        System.out.println("After remove(object): " + cities);

        // removeFirst()
        String first = cities.removeFirst();
        System.out.println("Removed first element: " + first);
        System.out.println("After removeFirst(): " + cities);

        // removeLast()
        String last = cities.removeLast();
        System.out.println("Removed last element: " + last);
        System.out.println("After removeLast(): " + cities);

        // offer(E e)
        cities.offer("Pune");
        System.out.println("After offer(): " + cities);

        // peek()
        System.out.println("Head using peek(): " + cities.peek());
        System.out.println("List after peek(): " + cities);

        // poll()
        System.out.println("Removed head using poll(): " + cities.poll());
        System.out.println("List after poll(): " + cities);
    }
}
```

Completion and Execution Commands

1. Save the program as:
`LinkedListMethodsDemo.java`
2. Open Command Prompt / Terminal and move to the folder containing the Java file.
3. Compile the program:
`javac LinkedListMethodsDemo.java`
4. Execute the program:
`java LinkedListMethodsDemo`
5. The results of all LinkedList operations are displayed on the console.

Output

```
After add(): [Hyderabad, Chennai, Mumbai]
After addFirst(): [Delhi, Hyderabad, Chennai, Mumbai]
After addLast(): [Delhi, Hyderabad, Chennai, Mumbai, Kolkata]
Element at index 2: Chennai
First element: Delhi
Last element: Kolkata
After remove(index): [Delhi, Hyderabad, Mumbai, Kolkata]
After remove(object): [Delhi, Hyderabad, Mumbai, Kolkata]
Removed first element: Delhi
After removeFirst(): [Hyderabad, Mumbai, Kolkata]
Removed last element: Kolkata
After removeLast(): [Hyderabad, Mumbai]
After offer(): [Hyderabad, Mumbai, Pune]
Head using peek(): Hyderabad
List after peek(): [Hyderabad, Mumbai, Pune]
Removed head using poll(): Hyderabad
List after poll(): [Mumbai, Pune]
```

Result

Thus, the Java program was successfully implemented and executed to demonstrate the commonly used methods of the LinkedList class, including list operations and queue operations.